Project 3
Game Theory-Based Multi-Kingdom Simulation

The Council of Monarchs – A Strategy for Peace

1 Project Description

This project aims to design a large-scale, game theory-based simulation where eight different kingdoms interact over 50 years. Each kingdom has a unique strategy, and these strategies determine how inter-kingdom relations evolve over the years. The simulation has a multi-layered structure involving cooperation–defection dynamics, trust updates, alliance formation, random probabilities, information decay (gossip), and a matching network. The entire process runs automatically, and strict adherence to technical rules is mandatory, as the accuracy of the output will be evaluated by an automated testing system.

Every year, all kingdoms meet each of the other seven kingdoms once. In each encounter, they secretly make a move: “C” (Cooperate) or “D” (Defect). These moves are scored according to the Prisoner’s Dilemma payoff matrix, and the kingdom’s annual total score is determined. The behavior of the kingdoms is not solely dependent on their strategic decisions; it is also influenced by moves played in previous years, their memory records, the reliability of their rivals, and the continent’s overall atmosphere. Each kingdom’s annual cooperation rate is calculated, which then reflects in a trust score, building up a reputation that increases or decreases over time. These trust scores play a decisive role both in forming alliances and in specific advanced strategies (especially Opportunist and Adaptive).

There are two types of randomness in the simulation: envoy error and palace gossip. Envoy error is the probability that the transmitted move is randomly reversed, affecting only that year’s encounter. Palace gossip occurs before the start of a new year, causing one kingdom’s actions from the previous year to be randomly perceived as "all C" or "all D" in the memory of all other kingdoms. This introduces fragility to the strategic balance, as it can corrupt long-term trust relationships.

Kingdoms are also subject to a dynamic alliance system. Kingdoms whose trust exceeds a certain threshold can form alliances with other suitable kingdoms at the start of the year. Alliances are determined by a mutual preference ranking system; both parties must choose each other at the exact same rank level. Formed alliances last for 5 years, during which time the parties are compelled to always cooperate with each other. If trust drops, an alliance can be dissolved prematurely. Alliances lead to the formation of peace clusters on the continent and strengthen long-term cooperation tendencies.

To run the simulation, all kingdoms and continental statistics are held in specific data structures. Kingdoms’ memories are dictionary structures that store past moves, and functions like `CoopRate` and `GeneralTrust` update based on this data. Continental data includes annual statistics like mean trust, mean cooperation, and cooperation variance. These statistics are directly used in the decision mechanisms of strategies like Adaptive and Opportunist. At the end of each year, the `ContinentData` function refreshes these values and plays a role in determining the next year’s trust thresholds.

All these structures are controlled by functions that the student is required to write as part of the project. Each function performs a specific task: decision-making, memory updating, trust calculation, error simulation, gossip processing, alliance formation and dissolution, encounter execution, and managing the annual cycle. The main simulation loop is executed in the `main` function, where gossip, alliance decay, diplomacy, all encounters, and statistical updates are processed each year. Since the format, content, and structure of the data structures and all outputs are verified by the testing system, strict compliance with the rules specified in the document is mandatory. Otherwise, even if the code works correctly in the local environment, incompatibility with the grader may result in a zero score.

This project expects the student not only to write correct algorithms but also to model a complex system in a consistent, organized, and error-free manner. This multi-layered structure, which combines kingdom behaviors, information flow, random errors, and strategic interactions, offers an opportunity to experience the dynamics of game theory in a software simulation environment.

2 Scenario Summary

The year is 1264. After centuries of warfare, eight major kingdoms on the old continent have signed a Treaty of Peace. This treaty is renewed every spring by the *Council of Monarchs*.

At the Council meeting each year, the kingdoms enter into bilateral agreements. In each agreement, the parties secretly make one of two decisions:

- **C (Cooperate):** Share resources for common defense and trade.
- **D (Defect):** Hoard resources, strengthening only their own kingdom.

The scoring is as follows:

Choice	Kingdom 1 Score	Kingdom 2 Score
(C, C)	3	3
(C, D)	0	5
(D, C)	5	0
(D, D)	1	1

This process lasts for 50 years (50 rounds). At the end of each year, the Council determines the winners and losers.

3 Kingdoms and Their Strategies

Kingdom	Monarch	Strategy
Arvendon	King Edric	Tit-for-Tat
Veloria	Queen Seren	Always Cooperate
Dornhal	Lord Garrick	Always Defect
Caldria	Duke Taren	Grudger
Elyndra	Prince Kael	Forgiver
Morenth	Knight Varyn	Random
Norvale	Count Roderic	Opportunist
Isador	Sage Elaria	Adaptive

4 Kingdom Decision Behaviors

- **King Edric (Tit-for-Tat):** A fair and principled ruler. Plays C in the first round, then reflects the opponent's last move in subsequent years. Responds to friendship with friendship, and to betrayal with betrayal; measures justice with balance.

- **Queen Seren (Always Cooperate):** An unconditional advocate for peace; always plays C, even if betrayed in the past. May be exploited by the ill-intentioned but ensures the continuation of order.

- **Lord Garrick (Always Defect):** An ambitious, ruthless, and distrustful ruler. Always plays D; wins in the short term, but earns the hatred of everyone in the long run.

- **Duke Taren (Grudger):** An honorable but stubborn noble. Plays C initially, but if deceived once, plays D against that person forever. Gains loyal friends but does not forgive.

- **Prince Kael (Forgiver):** A kind yet strategic ruler. Plays C in the first round. If betrayed, plays D in the following year as a warning, then monitors the opponent's attitude. If the opponent plays C for two consecutive years, Kael forgives and reverts to C. This cycle allows a chance for errors to be rectified, but Kael remains cautious in situations of continuous betrayal.

- **Knight Varyn (Random):** Chaotic and unpredictable. Leaves decisions entirely to chance each year; sometimes builds friendship, sometimes breaks the peace. Has no rational strategy, is merely on the side of probability.

  Behavior Model:

  $$S_i(t+1) = \begin{cases} C, & \text{with probability } p_C \\ D, & \text{with probability } 1 - p_C \end{cases}$$

  Here p_C is the probability that Knight Varyn cooperates. By default, $p_C = 0.5$.

- **Count Roderic (Opportunist):** Counts the moves of not only his enemies but also his friends in his court. Roderic measures trust not only from his own experience but also from the behavior of other kingdoms towards the entire continent. Roderic defaults to playing C for the first year.

  Decision Rule:

  $$S_i(t+1) = \begin{cases} C, & g_j(t) > \theta_g(t) \text{ and } r_j(t) > r_i(t) \\ D, & \text{otherwise} \end{cases}$$

  Variables:

  - $S_i(t+1, j)$ – Kingdom i's decision against kingdom j in year $(t+1)$ (**C** or **D**).
  - $g_j(t)$ – Kingdom j's general trust score at the end of year t $(0 \leq g_j(t) \leq 1)$. This value is updated according to the system defined in the "General Trust Score and Cooperation Rate Mechanism" section.
  - $\theta_g(t)$ – The dynamic trust threshold, updated annually based on the continent's general trust (initially set as $\theta_g(0) = 0.2$).
  - $r_j(t)$ – Kingdom j's total payoff (score) in year t.
  - $r_i(t)$ – Roderic's own total payoff in year t.

Dynamic Trust Threshold Update:

$$\theta_g(t+1) = (1-\beta)\,\theta_g(t) + \beta\,\mu_g(t)$$

Where:

- $\mu_g(t) = \dfrac{1}{N}\sum_{i=1}^{N} g_i(t)$ – The continent's mean trust level in year t.
- $\beta = 0.3$ – Roderic's *adaptation coefficient*, determining his sensitivity to the general atmosphere.

- **Sage Elaria (Adaptive):** Elaria's goal is not profit, but maintaining stability. She senses whether the continent's order is being disrupted by monitoring the variation in kingdoms' cooperation rates. She disregards small fluctuations; however, if there is a significant *decrease* in cooperation variation, she strengthens cooperation (**C**), but if there is a significant *increase*, she withdraws her trust (**D**), believing order is breaking down. If the change is below the δ threshold, Elaria maintains her current behavior.

  Decision Rule:

$$S_i(t+1) = \begin{cases} C, & \sigma_C(t) < \sigma_C(t-1) - \delta, \\ D, & \sigma_C(t) > \sigma_C(t-1) + \delta, \\ S_i(t), & |\sigma_C(t) - \sigma_C(t-1)| \le \delta. \end{cases}$$

  Variables:

  - $\overline{C_i}(t)$ – Kingdom k's cooperation rate in year t.
  - $\mu_C(t) = \frac{1}{N}\sum_{i=1}^{N} \overline{C_i}(t)$ – The continent's average cooperation rate.
  - $\sigma_C(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left(\overline{C_i}(t) - \mu_C(t)\right)^2}$ – A measure of the cooperation variation among kingdoms (standard deviation).
  - $\delta = 0.05$ – *Balance tolerance*; small fluctuations are ignored.
  - $N = 8$ – Total number of kingdoms.

  Since there is no historical data in the first year, Elaria starts with **C**. The change also does not exceed the δ threshold in the second year, so Elaria maintains her behavior. In this model, Elaria is the protector of *balance*, not peace: she withdraws her trust when stability is disrupted, but remains patient with small deviations.

5 General Trust Score and Cooperation Rate Mechanism

The behavior of each kingdom is shaped not only by its strategy but also by the level of cooperation it has exhibited in the past. For this purpose, a general trust score $g_i(t)$ and an annual cooperation rate $\overline{C_i}(t)$ are calculated for each kingdom.

5.1 Annual Cooperation Rate

A kingdom's cooperation rate in year t is calculated based on how many of its total 7 agreements in that year were a "C" move:

$$\overline{C_i}(t) = \frac{\text{Number of agreements played as C}}{N-1}$$

This rate is a value between 0 and 1:

- 1: The kingdom played **C** in all encounters that year.
- 0: The kingdom played **D** in all encounters that year.
- 0.5: The kingdom maintained a balance between cooperation and defection.

5.2 General Trust Score Update

Each kingdom's reputation, i.e., its trust score, is updated with a balance between the previous year's average and this year's behavior:

$$g_i(t+1) = g_i(t) + \alpha(\overline{C_i}(t) - g_i(t))$$

Where:

- $g_i(t)$ – The kingdom's current trust score (in the range 0–1),
- $\overline{C_i}(t)$ – The cooperation rate for that year,
- $\alpha = 0.25$ – The update coefficient, determining how quickly trust changes.

Thanks to this formula, each kingdom's trust score changes dynamically; kingdoms that cooperate regularly increase their reputation, while those that constantly defect decrease theirs.

Note: The trust score must remain between 0 and 1. Keeping it within this boundary is important for both strategic balance and the stability of the alliance system.

Table 1: Initial Trust Scores ($g_j(0)$)

Monarch	$g_j(0)$	**Brief Description**
King Edric	0.6	Fair and balanced.
Queen Seren	0.7	Always cooperative.
Lord Garrick	0.3	Untrustworthy, ambitious.
Duke Taren	0.5	Loyal but grudging.
Prince Kael	0.55	Forgiving and cautious.
Knight Varyn	0.45	Random, unpredictable.
Count Roderic	0.5	Opportunistic, analytical.
Sage Elaria	0.5	Neutral observer.

6 Matching System and Randomness

The structure of mutual interaction among the kingdoms is shown in Figure 1. This network represents the possible agreements each kingdom can make with the others.

- Each year, every kingdom is matched with the other 7 kingdoms once; thus, there are a total of $(8 \times 7)/2 = 28$ encounters each year.
- All kingdoms decide simultaneously each year; thus, the matches occur concurrently within the same year. Each kingdom sends C or D moves against the other 7 kingdoms at the same time, and the results are evaluated at the end of the year.
- Each encounter is an agreement, and C or D is played according to the strategies.
- Scores are calculated according to the payoff table and summed for the year-end scores.

6.1 Randomness Factors

1. **Envoy Errors (ε):** An error may occur during the transmission of decisions; with probability ε, C $\leftrightarrow$ D, and D $\leftrightarrow$ C are reversed. $\varepsilon = 0.05$ is used; meaning a communication error occurs in 5% of the decisions. (This is a technical communication error.)

2. **Palace Gossip (Memory Corruption):** Before the start of a new year, a rumor spreads about a random kingdom with a 10% probability. This rumor only corrupts the perception of the moves from the past year: all other kingdoms forget what that kingdom played last year and remember its moves against all rivals for the previous year as either entirely C or entirely D. The choice is made randomly and is the same for everyone.

3. **Alliances (Alliance System):** At the beginning of the Council meeting each year, there is a 20% probability that new alliances may arise between the kingdoms. This rate indicates that some of the total $(8 \times 7)/2 = 28$ possible relationship links between the 8 kingdoms may transition into an "alliance" state. An alliance is an agreement between two kingdoms based on mutual trust and lasts for 5 years. A kingdom can only be in one alliance at a time.

   Alliance Selection: Alliances are not entirely random; they are determined by trust scores and a small chance factor:

   (a) At the start of the year, all 28 possible kingdom pairs are generated.

   (b) For each pair (i, j), the following conditions are checked:

      i. Each kingdom pair (i, j) is considered a candidate link.

      ii. If there is already an existing alliance between the pair (i, j), this link is disregarded.

      iii. If *either* of the two kingdoms is part of another alliance, the calculation is not performed for this link. A kingdom can only be in one alliance at a time (Figure 2).

      iv. For the remaining links (where neither kingdom has an existing alliance), the following conditions are evaluated:

         - The trust score of both kingdoms must meet the threshold:

$$g_i > \theta_{\text{trust}} \quad \text{and} \quad g_j > \theta_{\text{trust}}, \qquad \text{where } \theta_{\text{trust}} = 0.6$$

         - If the conditions are met, a probability draw is made for the link:

$$r = \text{random}() \in [0, 1]$$

         If $r < 0.20$, this link is established as an alliance link.

   (c) The same kingdom may be part of multiple candidates. In this case, a score is assigned to each candidate pair:

$$U_{ij} = w_t \times \frac{(g_i + g_j)}{2} + w_r \times random()$$

   Here g_i and g_j are the trust scores, w_t is the trust weight, and w_r is the randomness weight. $w_t = 0.7$, $w_r = 0.3$ are used.

   (d) Each kingdom creates a *preference ranking* for the pairs it is a candidate for, based on the utility score. Let $\pi_i = (j_1, j_2, \ldots)$ be the ranking of kingdom i; this ranking is in descending order of U_{ij}. Also, let $\text{rank}_i(j)$ denote the rank number of j in i's ranking (1 is the best).

(e) Alliances are formed based on the principle of mutual same rank: For increasing values of $k = 1, 2, \ldots$, all still-unmatched pairs (i, j) satisfying the condition

$$\text{rank}_i(j) = k \text{ and } \text{rank}_j(i) = k$$

are matched in the same round. Matched kingdoms are removed from the list, and the next step k is proceeded to. If a mutual same rank cannot be found for any k, the kingdom remains unallied for that year.

Alliance Effect: During an alliance, the two kingdoms always play C in all their encounters with each other. They continue to implement their own strategies against other kingdoms. This creates "peace clusters" in the system.

Alliance Decay: If the trust score of one of the kingdoms drops below the set threshold before the alliance duration expires, the alliance ends prematurely. In this case, both parties exit the alliance state and can be candidates again.

Alliance Duration and Renewal: The default duration is 5 years. The alliance automatically ends when its duration expires; no renewal possibility or extension is applied.

Difference Between Envoy Error and Palace Gossip: Envoy error is a technical glitch in message transmission and only affects that year. Palace gossip is a social information error that corrupts perception; it poisons memories at the beginning of the new year, shakes a kingdom's reputation, and leads to long-term trust crises.

7 Critical Technical Rules and Zero Score Reasons

This project is evaluated by an automated grader. Even the smallest deviation from the technical rules given in the document may lead to the project receiving a direct 0 score. Therefore, complete compliance with all rules is mandatory.

7.1 Function Parameter Order

The order of parameters given in the function prototypes cannot be changed. The testing system calls your functions in this order; a change in the ordering will cause input values to go to the wrong places.

7.2 Naming and Case Sensitivity

Python is case-sensitive. Function and variable names must be written exactly as specified in the document.

- `Decide` $\neq$ `decide` $\neq$ `DECIDE`
- `Continent` $\neq$ `continent`

Incorrect naming will result in a `NameError` in the grader.

7.3 Order of the `Monarchs` List

The `Monarchs` list must follow the order of the kingdoms given in the scenario. Since some tests access kingdoms by index number, a change in order will cause the tests to fail.

7.4 File Name Format

The name of the Python file you submit must be exactly in the following format:

`INF211_Proje3_SchoolNumber.py`

The grader searches for your file with this name. If a different naming is used (`Proje3.py`, `inf211.py`, `final.py`, etc.), the testing system will be unable to load your file, and the project will receive 0 points.

7.5 General Format Compliance

In coding, adherence to the basic structure and fields specified in the document is essential. The testing system checks for the existence of specific keys and data fields. Significant format errors in these fields may lead to the functions not receiving the expected inputs, resulting in a zero score.

8 Project Tasks and Analysis

Write the following functions *with these exact names*. You must adhere to the input and output formats of the functions. Apply the rules and formulas described in the preceding sections for the internal logic of the functions. Unless otherwise specified, round all decimal numbers to 3 decimal places.

8.1 Data Structures and Initial State

Before starting to code, you must completely define the following two data structures, which form the basis of the project.

`Continent` (Dictionary): A single dictionary that holds the general statistics of the continent. It must initially have the following keys and values:

- `"year"`: 0 (Simulation year)
- `"theta_g"`: 0.2 (Initial trust threshold)
- `"mean_trust"`: 0.5125 (Initial average continental trust)
- `"mean_coop"`: 0.0
- `"sigma_coop"`: 0.0
- `"sigma_prev"`: `None`

`Monarchs` (List): A list containing 8 kingdom dictionaries. Each kingdom dictionary must have the following schema:

- `"name"`: Monarch Name (e.g., `"Edric"`)
- `"strategy"`: Strategy name (e.g., `"tit_for_tat"`)
- `"trust"`: Initial trust score (taken from Table 1, e.g., 0.6)
- `"score"`: 0
- `"coop_rate"`: 0.0
- `"ally"`: `None`
- `"ally_years"`: 0
- `"memory"`: **An empty Dictionary (`{}`).**

↪ *Hint:* As the code runs, this dictionary will hold rival names as keys. The values should be a list of Tuples in the form of `(MyMove, OpponentMove)`.

↪ **Example (King Edric's Memory):**
`{"Seren": [('C', 'C'), ('C', 'C')], "Garrick": [('C', 'D'), ('D', 'D')]}`
(Note: Since Garrick always defected, Edric responded to him with D in the second round.)

8.2 ContinentData(monarchs, continent)

Input: `monarchs`, `continent`.
Output: None (in-place update).
Task: Calculate the continent's average trust (`mean_trust`), average cooperation (`mean_coop`), and cooperation standard deviation (`sigma_coop`) using the data in the `monarchs` list and save them to the `continent` dictionary. Remember to store the previous variance in the `sigma_prev` field.

8.3 TrustThreshold(continent, beta)

Input: `continent`, `beta`.
Output: New `theta_g` value (float).
Task: Calculate the new θ_g value according to the "Dynamic Trust Threshold" formula, using the continent's current average trust, update the dictionary, and return the new value.

8.4 GeneralTrust(monarch, alpha)

Input: `monarch`, `alpha`.
Output: New `trust` value (float).
Task: Update the trust score (g_i) of the relevant kingdom using its current cooperation rate (see General Trust Score Update formula).

8.5 CoopRate(monarch)

Input: `monarch`.
Output: `coop_rate` (float).
Task: Scan all past moves in the kingdom's memory (`memory`), calculate and record the ratio of its own played "C" moves to the total number of moves.

8.6 Decide(strategy, hist, my_score, opp_score, opp_trust, Continent)

Input: Strategy name and game parameters.
Output: `'C'` or `'D'`.
Task: Implement the logic corresponding to the given `strategy` name (see "Kingdom Decision Behaviors" section) to determine the next move. Use `Continent` data for complex strategies like `Opportunist` and `Adaptive`.

8.7 EnvoyMis(move, rate)

Input: `move`, `rate`.
Output: Move (str).
Task: With the given `rate` probability, reverse the incoming move (C $\leftrightarrow$ D). Otherwise, return the move as is.

8.8 PalaceGossip(monarchs, gossip_rate, year, verbose)

Input: Simulation data and parameters.
Output: `True` (if a change occurred) or `False`.
Task: Implement the "Palace Gossip" rules. If the probability occurs, randomly select a target kingdom and change the perception of its past year's moves in the memories of other kingdoms to a fake move.

Verbose Mode (`verbose=True`): If gossip occurs and `verbose` is enabled, print the following information to the console in this format:

- `"[GOSSIP <Year>] <TargetMonarch> perceived as '<FakeMove>' by <AffectedCount> others"`

8.9 AllianceDecay(monarchs, theta_trust, verbose)

Input: `monarchs`, `theta_trust`, `verbose`.
Output: None (in-place update).
Task: Decrease the duration of existing alliances by 1 year. Dissolve alliances mutually that have expired or where the trust of one party has dropped below the θ_{trust} threshold.

Verbose Mode (`verbose=True`): When an alliance is dissolved, print the following information to the console in this format:

- `"[DECAY] Alliance between <Monarch1> and <Monarch2> dissolved (<Reason>)."`
- *Note: The reason should specify `"expired"` (duration ended) or `"trust drop"` (trust dropped).*

8.10 Diplomat(monarchs, theta_trust, wt, wr, p_gate, verbose)

Input: Monarchs and weight parameters.
Output: List of formed alliances `[(A,B), ...]`.
Task: Identify eligible pairs without an alliance. Implement the U_{ij} scoring and ranking algorithm from the "Alliance Selection" section to form and record new alliances.

Verbose Mode (`verbose=True`):

- If new alliances are formed:
  `"[DIPLOMAT] New alliances formed:  [('MonarchA', 'MonarchB'), ...]"`
- If no alliances are formed:
  `"[DIPLOMAT] No new alliances this year."`

8.11 Council(mon1, mon2, Monarchs, Continent, envoyMisRate, verbose)

Input: The two monarchs to meet and general data.
Output: Tuple `(m1, m2, p1, p2)`.
Task: Match the two kingdoms. If an alliance exists, force automatic cooperation; otherwise, determine the move with the `Decide` function. Add an error margin with `EnvoyMis`, calculate and add the scores, and record the final moves in the memory (`memory`).

Verbose Mode (`verbose=True`): When each encounter process is completed, print the following to the console in this format:

- `"[COUNCIL] <Monarch1> vs <Monarch2> -> (<M1>, <M2>) -> +<Score1>, +<Score2>"`

8.12 main(years, envoy_rate, gossip_rate, verbose)

Input: Simulation parameters.
Output: `(Monarchs, Continent)`.
Task: Set up the annual simulation loop. Each year, process the following steps in order: Gossip, Alliance Decay, Diplomacy, Council (for all pairs), Statistical Updates.

Verbose Mode (`verbose=True`): Provide information to the console at specific stages of the loop:

- At the start: `"Simulation begins: <Year> years total"`
- At the start of each year: `"===== YEAR <Year> ====="`
- At the end of each year (Statistics):
  `"Avg Cooperation = <AvgC>, Avg Trust = <AvgT>, theta_g = <Theta>"`
- At the end of the simulation: Print the ranked list of monarchs (Score, Trust, Cooperation Rate).

9 Additional Information: What are C and D?

The Origin of the Abbreviations. In game theory, there are two fundamental actions:

- **C** = *Cooperate*: Act in mutual interest, engaging in trust-based sharing with the other party. This strategy increases peace and average payoff in the long run.
- **D** = *Defect*: Provides individual gain in the short term, but damages trust and reduces total welfare in the long run.

These two behaviors determine the balance within a society, especially in repeated games (e.g., the Prisoner's Dilemma). The essence of game theory is to show that individual rationality can lead to collective ruin.

Strategic Thought: The balance between cooperation and defection is related not only to self-interest but also to information accuracy and memory reliability. As misinformation (gossip) or communication error (envoy error) increases, the system's order is disrupted. This situation shows that strategic trust is mathematically fragile, even in the real world.

10 Visual Appendices

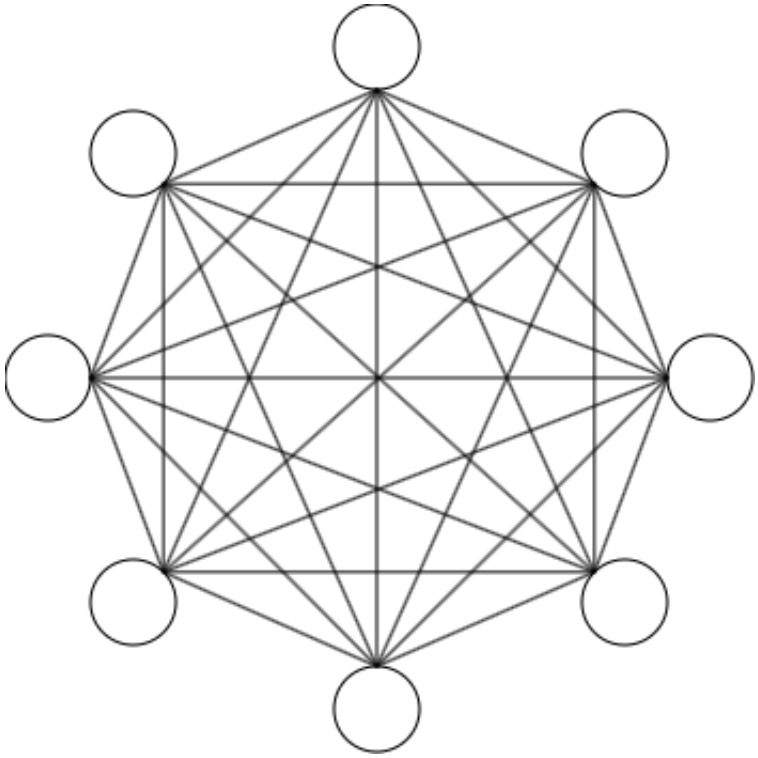

Figure 1: Network diagram of the eight kingdoms.

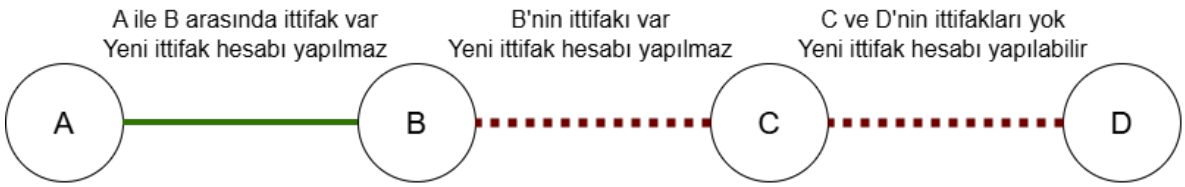

Figure 2: Example of network-based alliance control